

Prague University of Economics and Business
Faculty of Informatics and Statistics



Econometric analysis of the relationship
between vaccination against COVID-19 and
its occurrence in population

BACHELOR THESIS

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Autor: Roman Shaibaneev

Supervisor: Vladimír Holý

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Declaration of Authorship

I declare, that my bachelor thesis *Econometric analysis of the relationship between vaccination against COVID-19 and its occurrence in population* was totally made by me except the literature and materials mentioned in bibliography.

May 2023, Prague

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Roman Shaibaneev

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Abstract

Over the whole COVID-19 pandemic the efficiency of vaccination was doubted by a large group of people. It was even thought that they are more harmful for people who took them. In this bachelor thesis I try to investigate the relationship between given vaccine doses and two variables - the ratio of deaths and recovered from the disease and its death rate. I try to estimate this effects by developing an ARDL model to deal with autocorrelation. I found a significantly negative total effect of the vaccines on the variables, whereas a more detailed analysis showed negative and significant effect on the variables from vaccine doses given in recent 3 months and also 9 months up until year. Also, logarithmic transformation does not create drastic changes in estimates having a little better residuals.

Keywords

ARDL model, COVID-19, autocorrelation, stationarity

Abstrakt

Během celé pandemie COVID-19 byla účinnost očkování zpochybňována velkou skupinou lidí. Dokonce se mluvilo, že jsou škodlivější pro lidi, kteří se nechali očkovat. V této bakalářské práci se snažím prozkoumat vztah mezi dávkami vakcín proti koronaviru a dvěma proměnnými – poměrem úmrtí a uzdravených, a fatalitou nemoci. Chci odhadnout tyto vlivy pomocí ARDL modelu, abych vyřešil problém autokorelace. Našel jsem významně negativní celkový efekt vakcín na proměnné, zatímco podrobnější analýza ukázala negativní a významný vliv na proměnné z dávek vakcín za posledních 3 měsíce a také 9 měsíců zpátky až do roku. Logaritmická transformace také nezpůsobila dramatické změny v odhadech, a to s trochu lepšími rezidui.

Klíčová slova

ARDL model, COVID-19, autokorelace, stacionarita

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Abbreviations

ARDL Autoregressive Distributed Lag

ARDL-ECM Autoregressive Distributed Lag Error Correction Model

OLS Ordinary Least Squares

PP-test Phillips-Perron test

ADF-test Augmented Dickey-Fuller test

BG-test Breusch-Godfrey test

LM Lagrange Multipliers

ACF Autocorrelation function

BLUE Best Linear Unbiased Estimator

ARCH Autoregressive Conditional Heteroscedasticity

Introduction

Epidemic of COVID-19 caused by SARS-CoV-2 started in Wuhan, China, in November-December of 2019. It spread rapidly in China and on the other continents becoming a world pandemic. The first official case in Europe was registered in Northern Italy on 21st February 2020, but there are evidences suggesting that the pandemic hit Northern Italy around 3 weeks earlier the official date [Cerqua and Di Stefano, 2022]. In Czech Republic first three registered cases were on the 1st of March, and its total cumulative amount grew up to more than 4.5 mln. cases.

To fight the pandemic most of countries used restrictive methods so their medical staff and facilities would not overload, and excess mortality would not rise, what happened in Italy during the first wave [Scortichini et al., 2020]. These restrictions led to very high economic costs especially in developing economies compared to costs from deaths [Gagnon et al., 2023], so it created a huge debate on whether restrictions were actually needed. That was also the reason why countries such as Sweden did not impose strict restrictions on their economy. But also held the discussion that vaccines against COVID-19 would become the most efficient way to fight the disease. Two years ago estimated total costs on them for the next 5 years were about 150 bln. [Mishra, n.d.].

In this thesis I'm trying to examine the effect of vaccination in Czech Republic on two variables: the ratio of deaths and recoveries, and the ratio of deaths and cases, suspecting that vaccine doses affect both of them negatively. The first variable can be interpreted as some odds that an exposed person shall die and the second can be a fatality rate, even though, due to macro nature of the data those will be just approximates. For the sake of briefness, I'll call my first variable just *the ratio* and *fatality rate*, respectively, and for both of them I'll refer as *variables of interest*.

1. Data

In this section I'll explain my data source I used, data preparation process and variables I chose for the future analysis.

My main and only source of data I used here was the Ministry of Health of the Czech Republic website. I downloaded four data sets from its public database [MZČR, 2023]:

- on COVID-19 cases
- deaths from it
- recovered patients
- vaccinations table

First three had a very similar structure - person id as a key, date of the event, person sex and other irrelevant information. Vaccinations data set was a bit different - information on doses was aggregated by day, region and age group. I considered the events since 01.03.2020 till 08.02.2023. I changed all data on total weekly events for more precision and then used these series for the analysis. From them I created my variables of interest - the ratio of deaths and recoveries, and fatality rate. The first variable I created simply by dividing number of deaths by the number of recoveries from the same week, whereas the fatality rate was created as the number of deaths divided by the number of cases but 3 weeks ago as it was estimated on Australian data that in average people die in 18 days after getting infected [Marschner, 2021]. I dropped first three weeks due to zeros in beginning of the epidemic, having 151 weeks left in total. Also I divided number of doses by 1×10^7 just to make my further results much more readable. So my main data table looks as follows

Table 1.1: Data for models

<i>week</i>	<i>ratio</i>	<i>fatal</i>	<i>vac_12</i>
...	...	...	...
43	0.0192	0.0279	0
44	0.0165	0.0284	1.34×10^{-3}
45	0.0195	0.0261	3.50×10^{-3}
46	0.0159	0.0249	7.13×10^{-3}
47	0.0154	0.0149	8.67×10^{-3}
48	0.0201	0.0109	7.30×10^{-3}
49	0.0193	0.0155	8.09×10^{-3}
...	...	...	...

These two variables and their logarithmic transformation are depicted on Figures 1.1-1.4 with blue line corresponding to linearly reshaped cumulative number of vaccine doses. Obviously, both variables have very similar shape, since they are created from related indicators, nevertheless, it's interesting how vaccines affect them.

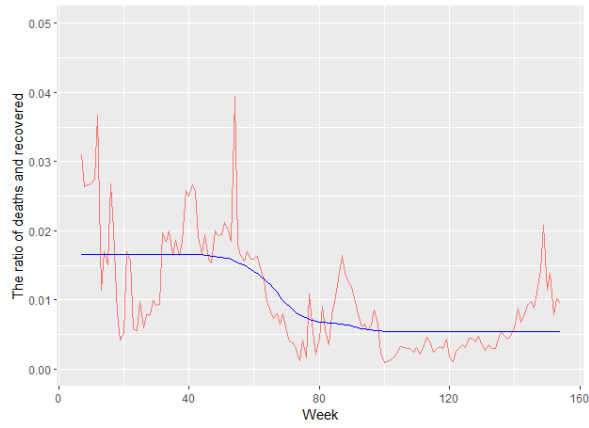


Figure 1.1: Level *ratio*

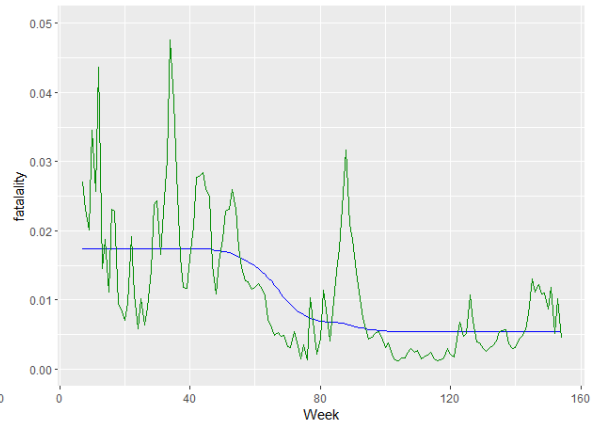


Figure 1.2: level *fatal*



Figure 1.3: log *ratio*

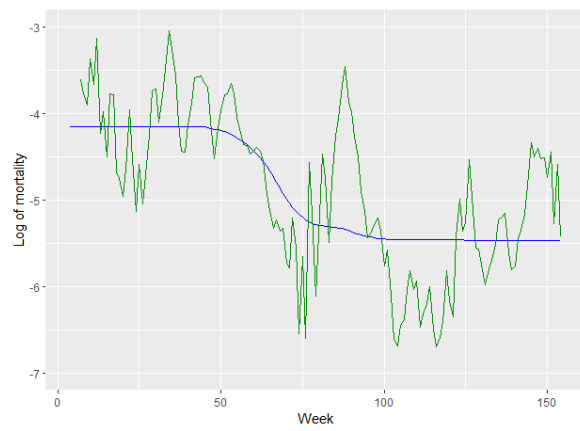


Figure 1.4: Log *fatal*

Visually, both variables seem to fall with vaccine doses rising, as it may be expected. Also the variables seem to lose in variance with vaccination, especially for levels. In this thesis I won't estimate this variance effect, but the observation is also interesting.

2. Methodology

Time series can be described as an ordered through time or some other dimension sequence of observations. The origin of time series analysis was to develop methods in forecasting, but it can also be used for understanding and description of the generating mechanism, and optimal control of a system. The intrinsic nature of these data is that they are correlated or even depended on their past or maybe future values, therefore the ordering of the observations is very important [Wei, 2006].

Theoretically, time series can be thought as a stochastic process, or a family of time indexed random variables $Z(\omega, t)$, where ω belongs to a sample space and t to an index set. The purpose of time series econometrics is to model such stochastic processes as an appropriate difference equation with random component and estimating it [Enders, 2014]. One of the biggest challenges in time series analysis is the presence of autocorrelation, which absence is one of the conditions for an OLS to be BLUE. Also, since my idea is to verify the effect of cumulative vaccine doses on my variables of interest I'll have to be dealing with variables of mixed integration, therefore I'd need a methodology that would be robust to my problem. That's why I used ARDL approach here [Shrestha and Bhatta, 2018].

2.1 Time series basic concepts

2.1.1 Ordinary Least Squares estimation

OLS is the most common method to estimate unknown parameters in a linear regression model:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + \epsilon_i \quad (2.1)$$

or

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \quad (2.2)$$

in matrix form, where $\mathbf{y}$ is a vector dependent variable, or regressand, $\mathbf{X}$ is a matrix of independent variables, $\boldsymbol{\beta}$ is a vector of parameters we want to estimate and $\boldsymbol{\epsilon}$ to be a vector of random disturbances.

Least Squares is a method that looks for a minimum of a function

$$\boldsymbol{\epsilon}^T \boldsymbol{\epsilon} = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \quad (2.3)$$

conditional on β . The method is said to be the best linear unbiased estimator when specific conditions are met. Those conditions are:

- **A1.** Linearity in parameters
- **A2.** Full rank of $\mathbf{X}$
- **A3.** Exogeneity
- **A4.** Homoscedasticity and no autocorrelation
- **A5.** Stochastic or nonstochastic data

Violation of these assumption can lead to various problems in estimates.

2.1.2 Autocorrelation

Since I am dealing with time series, autocorrelation is very common, so this assumption is the most violated in this type of data. It can be defined as

$$\text{Corr}(u_t, u_s | \mathbf{X}) \neq 0, \text{ for all } t \neq s \quad (2.4)$$

Violation of 2.4 leads to biased estimates of standard errors and to inefficiency of OLS. It can be dealt with by applying various methods that can be suitable for the case. Here I'll add AR lags, but other methods can be tried as well (i.e. using a robust estimator) [Baltagi, 2011].

In empirical work the term 2.4 is very hardly estimated, so we need to define stationarity and ACF with PACF.

2.1.3 Stationarity

Assuming that the index set is the set of all integers, we can consider a finite sample $\{Z_{t_1}, Z_{t_2}, \dots, Z_{t_n}\}$ from a stochastic process $\{Z(\omega, t) : t = 0, \pm 1, \pm 2, \dots\}$. The n -dimensional distribution function is then defined as follows

$$F_{Z_{t_1}, \dots, Z_{t_n}}(x_1, \dots, x_n) = P\{\omega : Z_{t_1} \leq x_1, \dots, Z_{t_n} \leq x_n\} \quad (2.5)$$

where $x_i, i = 1, \dots, n$ are real numbers. This process is said to be strongly or strictly stationary if

$$F_{Z_{t_1}, \dots, Z_{t_n}}(x_1, \dots, x_n) = F_{Z_{t_1+k}, \dots, Z_{t_n+k}}(x_1, \dots, x_n) \quad (2.6)$$

for any n -tuple $(t_1, \dots, t_n)$ and k of integers.

We can also suppress the variable ω in the term $Z(\omega, t)$ and simply write to a stochastic process as Z_t . Then a given real-valued process $\{Z_t : t = 0, \pm 1, \pm 2, \dots\}$ is weakly or covariance

stationary if its mean and variance functions are time irrelevant and covariance function is a just a function of time gap between points, or the mean function is

$$E(Z_t) = \mu_t = \mu \quad (2.7)$$

the variance function

$$E(Z_t - \mu_t)^2 = E(Z_t - \mu)^2 = \sigma_t^2 = \sigma^2 \quad (2.8)$$

and the covariance function between Z_{t_1} and Z_{t_2}

$$E(Z_{t_1} - \mu_{t_1})(Z_{t_2} - \mu_{t_2}) = E(Z_{t_1} - \mu_t)(Z_{t_2} - \mu_t) = \gamma(t_1, t_2) = \gamma(t, t + h) = \gamma_h \quad (2.9)$$

If 2.7 and 2.8 exist, then strict stationarity implies weak stationarity, but not otherwise [Wei, 2006].

2.1.4 ACF and PACF

For a stationary process a autocorrelation function can be defined as

$$\rho_h = \gamma_h / \gamma_0 \quad (2.10)$$

where γ_0 is variance. For a stationary process its values must be around 0, except for the values of ρ_0 . Their graph is called a correlogram [Enders, 2014]. The most we are interested in a correlogram of residuals.

The main property of the ACF is that some term ρ_s , $s > 1$, can be nonzero even if the term Z_{t-s} does not appear in data generating process of Z_t , but it indirectly affects it through the terms between them. That's why a partial autocorrelation can be defined, which takes into account only direct effects and eliminates effects of Z_{t-s} on Z_t from the terms Z_{t-1} through Z_{t-s+1} . The PACF can be gotten from the regression

$$Z_t = \phi_{s1}Z_{t-1} + \phi_{s2}Z_{t-2} + \dots + \phi_{ss}Z_{t-s} + u_t \quad (2.11)$$

The partial autocorrelation function between Z_t and Z_{t-s} is the term ϕ_{ss} [Enders, 2014].

In empirical work we want values of residual ACF and PACF to be insignificant, so they imitate the White noise process.

2.1.5 BG-test for serial correlation

Breusch-Godfrey test uses Lagrange multiplier form statistic to check for q order serial correlation. It takes residuals $\hat{\epsilon}_t$ from the model and builds an auxiliary regression of the residuals on their lags and independent variables $\mathbf{x}_t$ used in the main model, and calculates the LM statistic as

$$\hat{\epsilon}_t = \rho_1 \hat{\epsilon}_{t-1} + \rho_2 \hat{\epsilon}_{t-2} + \dots + \rho_q \hat{\epsilon}_{t-q} + \boldsymbol{\lambda}^T \mathbf{x}_t + e_t \quad (2.12)$$

$$H_0 : \rho_1 = \rho_2 = \dots = \rho_q = 0$$

$$H_1 : nonH_0$$

$$LM = (n - q)R_{\epsilon}^2 \sim \chi_q^2 \quad (2.13)$$

where ρ_i is an autocorrelation of order i and R_{ϵ}^2 is the coefficient of determination from auxiliary regression 2.12 [Wooldridge, 2019].

2.1.6 ARCH LM-test

ARCH test for heteroscedasticity of order h takes residuals from the main model and builds and auxiliary regression of order r as

$$\hat{\epsilon}_t^2 = b_0 + b_1 \hat{\epsilon}_{t-1}^2 + b_2 \hat{\epsilon}_{t-2}^2 + \dots + b_h \hat{\epsilon}_{t-h}^2 + e_t \quad (2.14)$$

and calculates the LM statistic as

$$H_0 : b_1 = b_2 = \dots = b_r = 0$$

$$H_1 : nonH_0$$

$$LM = nR_{\epsilon^2}^2 \sim \chi_h^2 \quad (2.15)$$

where $R_{\epsilon^2}^2$ is the coefficient of determination from auxiliary regression 2.14 [Catani and Ahlgren, 2017]. The null hypothesis states that there is no heteroscedasticity, so we do not want to reject it.

2.1.7 Autoregressive Distributed Lag Model

The ARDL($p, q_1, q_2, \dots, q_k$) model is specified as follows

$$\Phi(L, p)y_t = \boldsymbol{\delta}^T \mathbf{w}_t + \sum_{i=1}^k \beta_i(L, q_i)x_{i,t} + u_t \quad (2.16)$$

where

$$\begin{aligned} \Phi(L, p) &= 1 - \sum_{i=1}^p \Phi_i L^i \\ \beta(L, q_j) &= 1 - \sum_{i=1}^{q_j} \beta_i L^i, \text{ for } j = 1, 2, \dots, k \\ \mathbf{u}_t &\sim iid(0; \sigma^2) \end{aligned}$$

L is a lag operator, $\mathbf{w}_t$ is a vector of deterministic variables. If we assume, that u_t is white noise, the ARDL model can be estimated consistently by OLS [Nkoro and Uko, 2016].

2.1.8 Augmented Dickey-Fuller test and Phillips Peron test

This test is used to find unit roots in a time series. It is a modification of a standard Dickey-Fuller test that takes higher order autocorrelation into account by considering a higher number of lags.

H_0 : There is a unit root

H_1 : $nonH_0$

The issue is that the test takes degrees of freedom to handle autocorrelation. That's why, a more robust to heteroscedasticity PP-test. This test makes a nonparametric correction to the t-test statistic, which handles higher order autocorrelations [Jalil and Rao, 2019].

3. Empirical analysis

3.1 Determining the order of integration

Since I'm having five variables, explanatory vaccine doses for each week, ratio between deaths and recovered, fatality rate and the logarithmic transformation of the last two, I'll conduct Phillips-Perron unit root test on each of the variables.

Variable	Order of integration
<i>vac_12</i>	1
<i>ratio</i>	0
<i>fatal</i>	0
$\log ratio$	0
$\log fatal$	0

Table 3.1: Determined order of integration

In Table 3.1 order for each variable is determined by PP-test. For my regressands I rejected the null hypothesis from first try and for vaccines I had to differentiate the series to reject the null of unit root of PP-test. From these results we can justify the usage of the ARDL approach.

3.2 Simple model with cumulative doses

Here I try to find a relationship between the variables of interest and total doses given at time t . I regress my variables of interest and their logarithmic transformations on their first lag and cumulative sum of vaccines for each week, which corresponds to a number of vaccine doses up to week t . This can be interpreted as fitting the blue line from Figures 1.1-1.4 in ARDL(1,0) model:

$$y_t = \alpha + \psi y_{t-1} + \beta \sum_{i=1}^t vac_12_i + \epsilon_t \quad (3.1)$$

where y_t is one of the four variables. The equation 3.1 is estimated with OLS and I fit four models. In Table 3.2 I report the results for this model. The estimates are very significant for both, the level and the log models.

Table 3.2: Comparison of estimates for *ratio*

Parameter	Model (regressand)			
	<i>ratio</i>	<i>fatal</i>	$\log ratio$	$\log fatal$
$\hat{\alpha}$	0.0089***	0.0100***	-1.1062***	-1.0661***
$\hat{\psi}$	0.4605***	0.4228***	0.7389***	0.7460***
$\hat{\beta}$	-0.0045***	-0.0051***	-0.2258**	-0.2413**

Note: *** p -value is less than 0.001

Note: ** p -value is less than 0.01

Note: * p -value is less than 0.05

Note: · p -value is less than 0.1

Figures 3.1-3.12 depict residual ACFs, PACFs, Breusch-Godfrey tests, residual plots and ARCH LM-test for each model. From those figures all models are quite acceptable. Unfortunately, ARCH LM-tests of first order fail for all models, so we reject homoscedasticity as it was maybe expected.

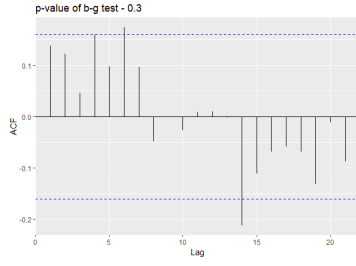
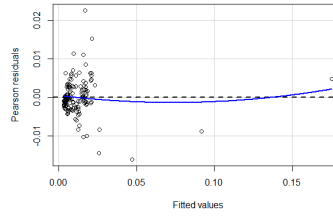


Figure 3.1: Residual ACF for *ratio* model



ratio model

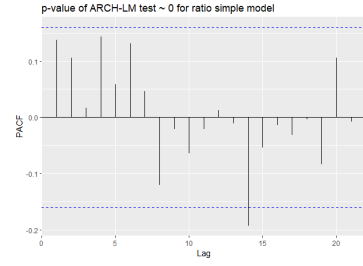


Figure 3.2: Residual PACF for *ratio* model

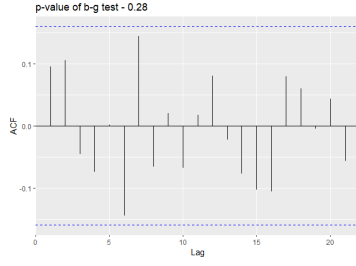
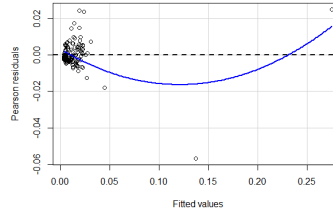


Figure 3.4: Residual ACF for *fatal* model



fatal model

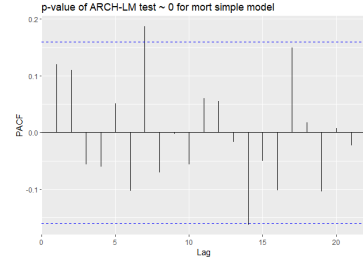


Figure 3.5: Residual PACF for *fatal* model

3.3 Models with different aggregations

In this section I'll try to change level of aggregation of the doses to investigate, how additional past information affect β estimates. That is, I fit a model

$$y_t = \alpha + \psi y_{t-1} + \beta \sum_{l=1}^k vac_12_{t-l} + \epsilon_t \quad (3.2)$$

where k is some point in past, indicating the week from which I start aggregation. Its minimum of one indicates the lowest aggregation - doses from the present week affect the present value of a regressand, while the highest aggregation - cumulative sum of all data, what I did in section 3.2. On Figures 3.13 and 3.15 those estimates for various k are depicted for level models, and on Figures 3.14 and 3.16 for logarithmic transformations. Colours indicate

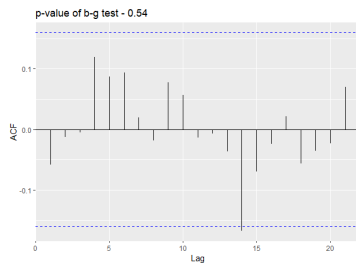
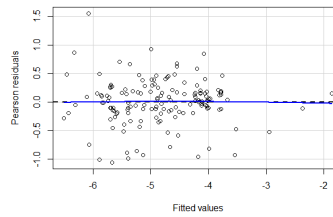


Figure 3.7: Residual ACF for *log ratio* model



log ratio model

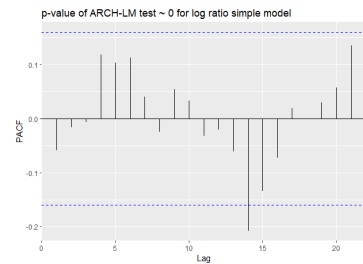


Figure 3.8: Residual PACF for *log ratio* model

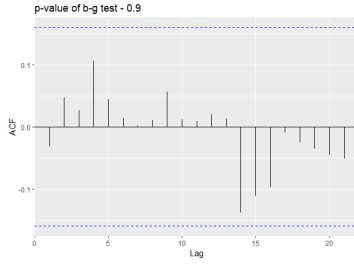


Figure 3.10: Residual ACF for log *fatal* model

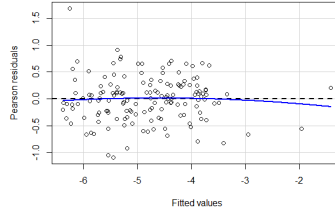


Figure 3.11: Residual plot for log *fatal* model



Figure 3.12: Residual PACF for log *fatal* model

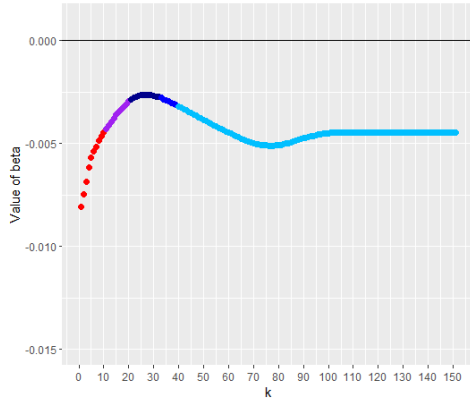


Figure 3.13: β estimates with various aggregations for *ratio* model

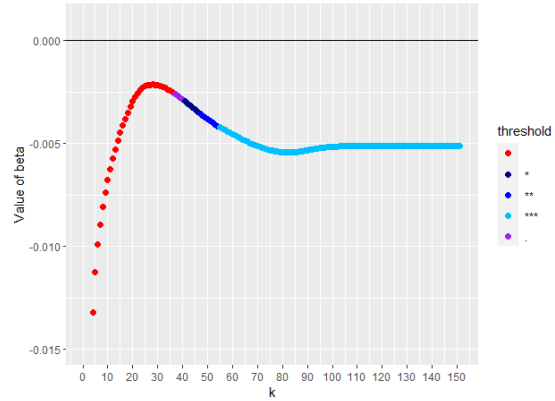


Figure 3.14: β estimates with various aggregations for *fatal* model

significance of the estimates the same as in Table 3.2.

These four plots have an interesting pathway: β estimates start being not significant, becoming more significant with higher values of k , find their peak on about 75-th week and then don't change after about 100-th week because the vaccination started 46 weeks after the epidemic outbreak. We can see how adding more weeks lowers the effect of considered doses on the dependent variables until k being higher than about 25. After that the effect rises, so now it's interesting to investigate this pattern closer.

3.4 Models with aggregation lags

Here I try to build models that aggregate some number of weeks of *vac_12* and also take doses before this aggregation into account as some other variables. Here I define two parameters, n and g , where n is the number of weeks I want to aggregate in each explanatory variable, and g is the number of historic groups, or explanatory variables I want to take into account. That is, if I choose n to be 10 for example and g to be 5 I take in total $g * n = 50$ weeks into account to explain my regressands, but in contrast to last part, I don't put 50 weeks into one variable, but into 5 variables by 10 each. The first variable being doses from the present week till 9 weeks ago, then the second variable consisting of the values from 10 weeks ago till the

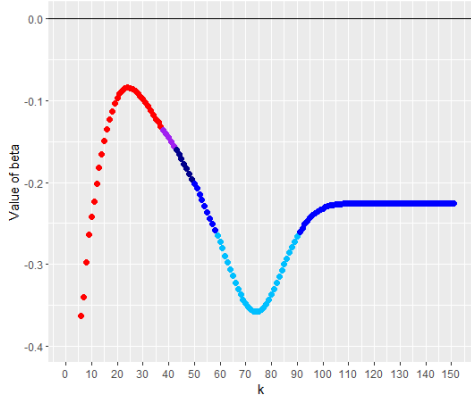


Figure 3.15: β estimates with various aggregations for log *ratio* model

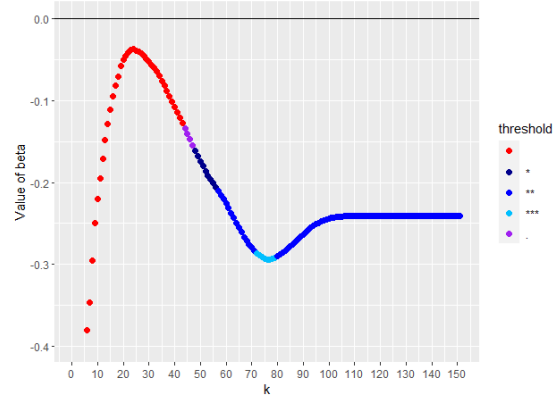


Figure 3.16: β estimates with various aggregations for log *fatal* model

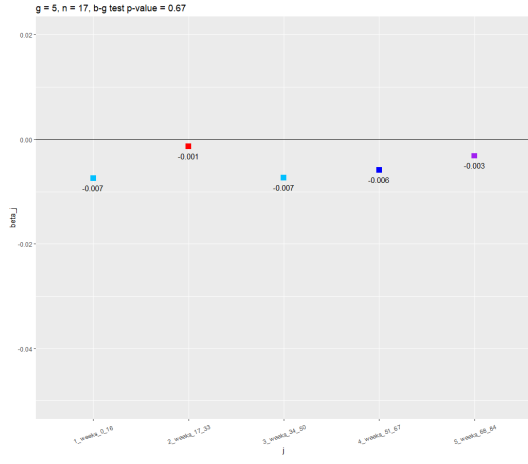
values 19 weeks ago, and so on. It can be considered as firstly aggregating doses by n weeks and then adding $g - 1$ lags, next being higher by n than the previous one. Mathematically the model can be represented as

$$y_t = \alpha + \psi y_{t-1} + \sum_{j=1}^g \beta_j \sum_{l=(j-1)n}^{jn-1} vac_12_{t-l} + \epsilon_t \quad (3.3)$$

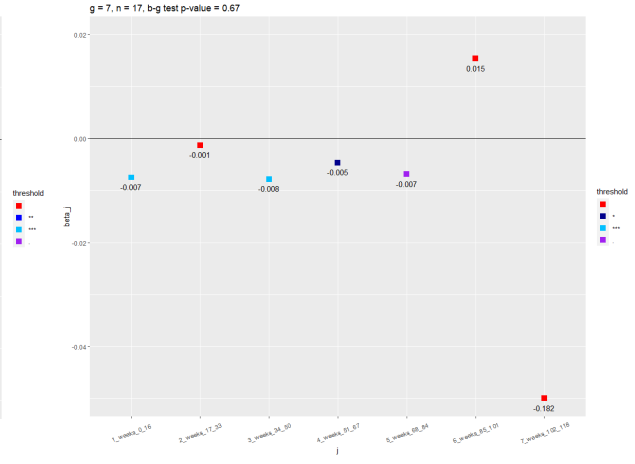
and before fitting it I require to choose n and g . Here I obviously want to diversify the effects as much as possible, but I do not want to overparameterize the equation. Also I need to use the results from the previous section and choose my total considered weeks in the model at least 50. Next I show some of the models on figures.

From the Figures 3.17 we can read that patterns of β_j coefficients are quite similar - there is a significant negative effect on the *ratio* of the first 3-4 months vaccinations, followed by insignificant effects for about half a year. Then a significant effect occurs again for a longer time, which can be caused because I did not distinguish between types of doses. After that the effects become less and less significant.

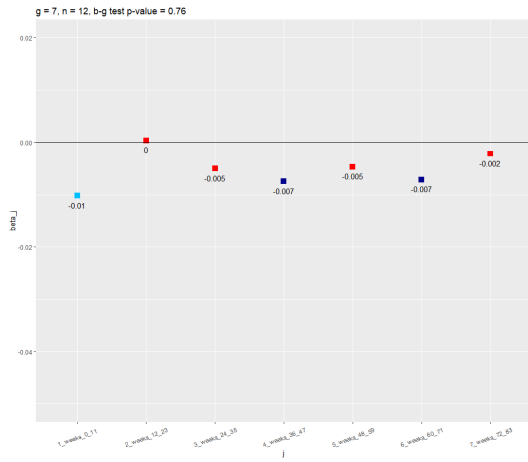
Effects of vaccines on level *fatal* have almost the same pattern as with *ratio* with the most effect from recent doses and the ones almost year ago. It can be captured comparing (a) and (b) subfigures with (c) and (d), where we can see that additional 5 weeks in recent vaccines lowers the effect and significance. The significance of vaccines almost year ago can be from second doses interacting with the first one.



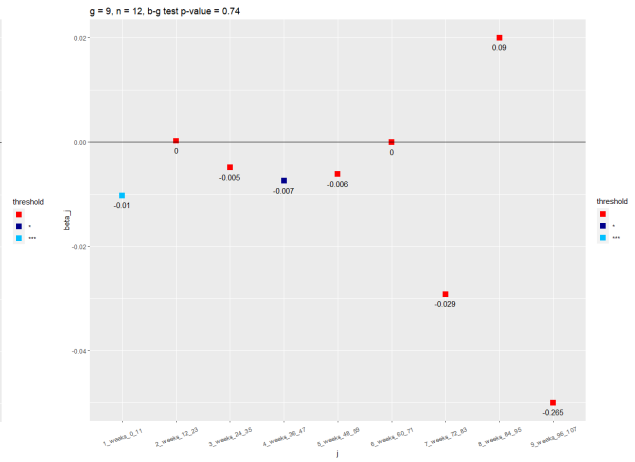
(a)



(b)

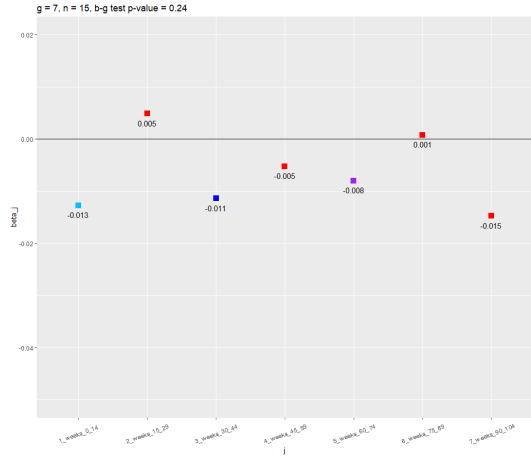


(c)

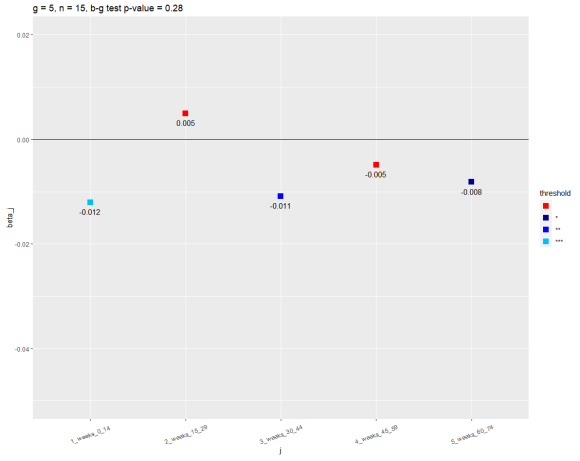


(d)

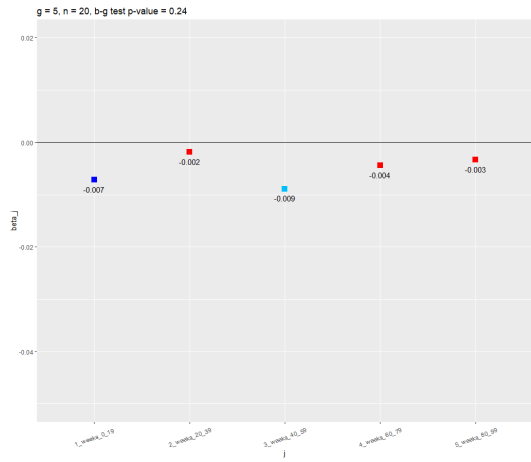
Figure 3.17: (a) Various models for *ratio*



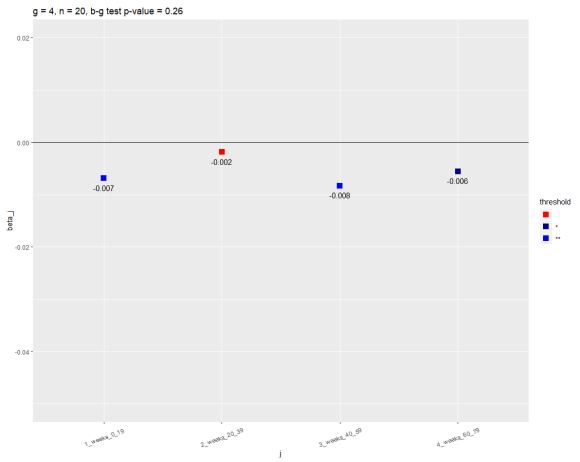
(a)



(b)



(c)



(d)

Figure 3.18: (a) Various models for *fatal*

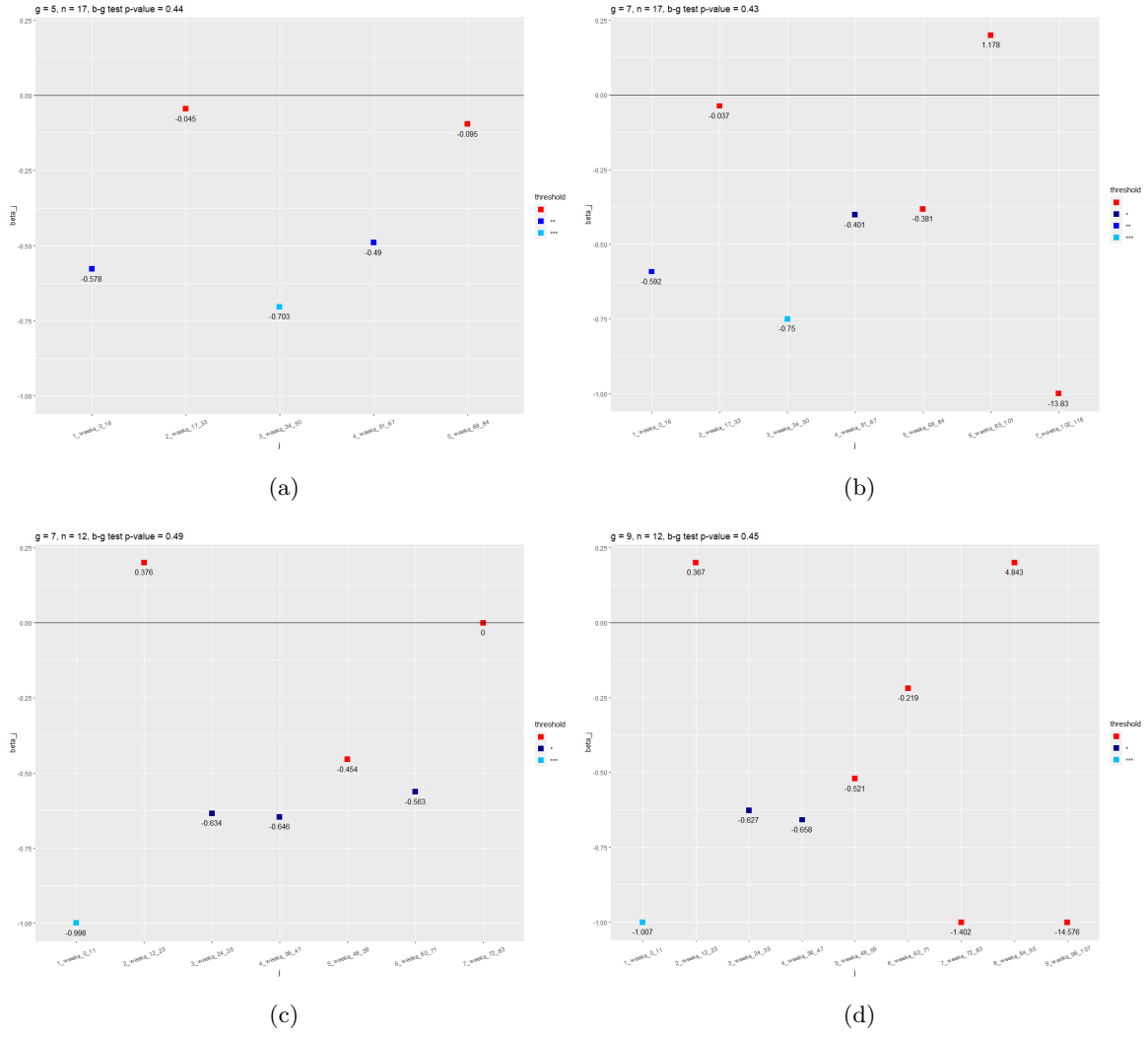
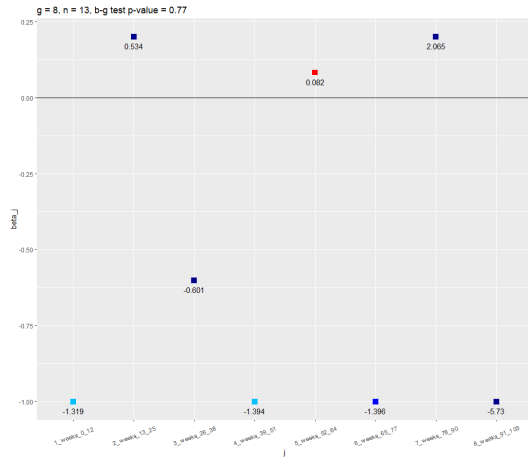
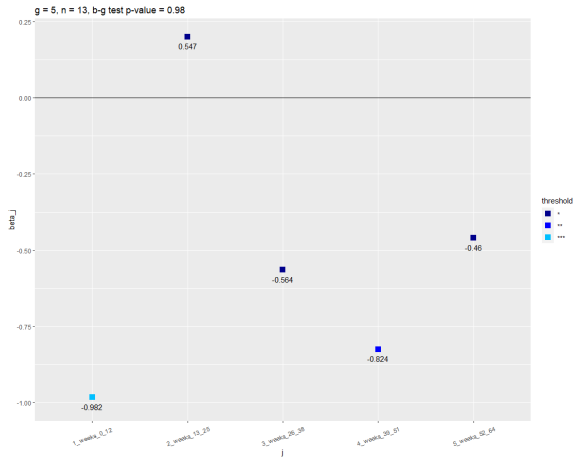


Figure 3.19: (a) Various models for $\log \text{ratio}$

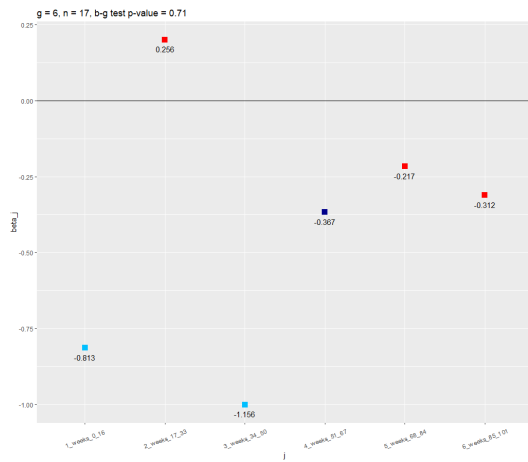
Very similar patterns with logarithmic regressands indicating no drastic changes between the models. Also, positive and sometimes significant values of estimates can explain why $\log \text{fatal}$ is the least significant on Figure 3.16 and ratio on Figure 3.13 is the most significant in the previous section.



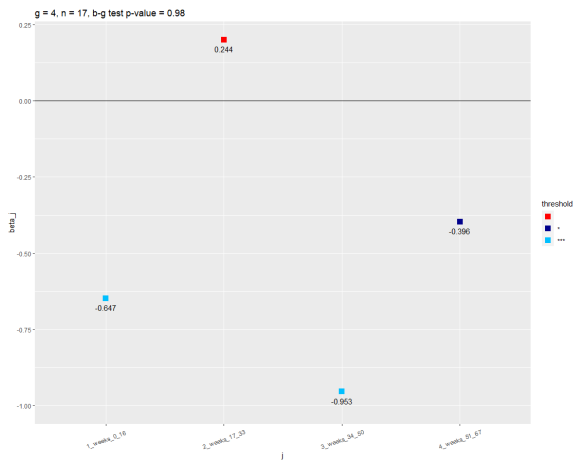
(a)



(b)



(c)



(d)

Figure 3.20: (a) Various models for log *fatal*

Discussion

In this part I'll try to point out the things that I didn't try here but think that can be crucial.

Firstly, the real data generating process is obviously much more complicated than my linear approach. There are different types of vaccines that interact with each other differently, there are many other variables that can be crucial for the analysis to take into account, like climate, government activity etc. Also, the true process might have nonlinear structure, since it is not possible to give infinite doses. My approach is just a try to catch relationships between the variables using a very simplified modelling.

Secondly, applying robust to autocorrelation estimators can be another possible way instead of using lagged regressands. This can catch the relationship even better. The reason I didn't use it was due to insignificant improvement in Durbin-Watson test for *ratio* simple model (3.1) by applying Cochrane-Orcutt estimator, so I didn't try to apply it on other models. But it is a possible way to look at the problem, which has its own advantages.

Conclusion

In this thesis I tried to capture the relationship of vaccination on two variables - the ratio of deaths and recoveries and death rate. I used weekly data of COVID-19 events, such as given vaccine doses, cases, deaths and recoveries for the analyses. I created both variables from these series and applied various ARDL models on it. A simple one, that regresses the rates on cumulative number of vaccines. Then I tried to capture how each additional week in cumulation affects the estimates. And lastly, tried to build a more complicated model which includes various aggregation lags to capture how recent weeks affect the regressands. Also I tried to apply same methodology on logarithmic transformation of dependent variables to make comparisons. Each model I controlled with AR(1) term to deal with serial correlation.

During the analysis I got following conclusions:

- Vaccination has a significant negative impact on the both dependent variables and their logarithmic transformations.
- There are weeks that can make the total effect of vaccination weaker, especially between 3 to 9 months ago.
- The main effect on the variables comes from recent vaccines, so they have almost instant impact on them.
- This effect does not hold long but comes back after 6-9 months, which can happen due to interaction with second and third doses of vaccine.
- Very similar results for logarithmic models excluding high changes due to transformation.

These results may not be perfect, due to the reasons I pointed out in discussion, but anyways are quiet interesting and maybe expected.

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